61Aa30A9 B893 42Fb B4Aa Cccb30Ba8160 Heterogenuous Multi Agent

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1. Overall Summary

This paper tackles the challenge of optimizing cooperative caching in Cognitive Radio Networks (CRNs), where dynamic content popularity and the interplay of cooperation and competition among users and Secondary Base Stations (SBSs) complicate traditional solutions. Existing DRL methods often fall short by treating the system as a single agent or homogeneous agents, or by operating in discrete action spaces. To address this, the authors propose a novel heterogeneous Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm. This approach uniquely models users and SBSs as two distinct types of agents, enabling them to autonomously learn to cooperate and compete to simultaneously minimize user content delivery delay and maximize SBS cache efficiency. A key innovation is allowing SBSs to optimize cached content fractions in a continuous space, offering greater flexibility than discrete eviction/retention. Leveraging centralized training and decentralized execution, the system reduces communication overhead. Numerical simulations demonstrate that the proposed heterogeneous MADDPG significantly outperforms homogeneous MADRL, DDPG, PPO, and TD3, achieving a substantial improvement in overall system reward by effectively balancing the mixed objectives of all agents.

2. Main Idea

The central concept is to propose a heterogeneous Multi-Agent Deep Deterministic Policy Gradient (MADDPG) approach for cooperative caching in Cognitive Radio Networks (CRNs). This approach treats both users and Secondary Base Stations (SBSs) as two distinct types of agents that interact, cooperate, and compete to simultaneously reduce user delivery delay and enhance SBS cache efficiency. It allows SBSs to optimize cached content fractions in a continuous action space and leverages centralized training with decentralized execution to reduce communication overhead.

3. Problem Being Solved

This paper addresses several limitations and research gaps in previous cooperative caching systems for Cognitive Radio Networks (CRNs). Existing approaches often fail to account for the autonomous nature of both users and cache servers, either by modeling the entire system as a single global agent or by treating multi-cache nodes as homogeneous agents. This leads to sub-optimal cache storage and content fetching strategies. Furthermore, many deep reinforcement learning (DRL) methods, like DQN, are limited to discrete action spaces, which restricts the flexibility of cache management (e.g., only evicting/retaining whole content). Centralized learning approaches also incur redundant communication overhead. The paper aims to solve the problem of dynamically optimizing cache storage efficiency and user content fetching strategies in a complex environment with heterogeneous agents that exhibit both cooperative and competitive behaviors.

4. Assumptions

- SBSs act as edge cache servers, proactively fetching hot contents from the primary base-station (PBS) during the off-peak time and serving users nearby through the secondary network.
- Users tend to fetch content from the cache node with the minimum delivery delay, while cache nodes expect to store popular contents to attract more users for maximum reputation or revenue.
- The slot duration To is long enough that all contents requests of users can be served within a time slot.
- The overall contents consist of F files, and the file size sf, f " F is independent identically distributed (i.i.d) of Pareto distribution with an average value of 5.
- Content request vectors qk(t) follow a Zipf distribution.
- Each file can be divided into fragments using a rateless code like Raptor codes.
- For user k, it can only retrieve one fragment of file f from SBS n with the size of An,f, and retrieve the rest fraction (1-An,f) from PBS, rather than employing multiple SBSs to cooperatively relay fragments.
- SBSs are geographically closer to users than PBS.
- The transmission rate of SBSs is faster than that of PBS (rn,k > ro,k).
- During training, global state and action (SAll, AAll) sharing requires 2(N+K) sets for information synchronization.

5. Limitations

6. Results and Conclusion

Experimental results confirm the effectiveness of the proposed heterogeneous MADDPG algorithm. The simulation, involving 5 SBSs and 3 users, shows that both user rewards (delivery delay minimization) and SBS rewards (cache efficiency maximization) converge stably during training episodes. The heterogeneous MADDPG significantly outperforms homogeneous MADDPG approaches, achieving a nearly 40% improvement in total system reward. Furthermore, it demonstrates superior performance compared to other DRL algorithms like DDPG, PPO, and TD3. The analysis reveals that SBSs adapt their caching strategies based on their geographic location and content popularity, with farther SBSs prioritizing hotter content more aggressively. The replacement cost factor also influences caching, where higher costs lead to a greater focus on popular content. The algorithm successfully balances the cooperative and competitive dynamics among heterogeneous agents, leading to reduced delivery delay for users and enhanced cache efficiency for SBSs.

7. Real World Impact

This research provides an effective DRL solution for optimizing content storage and delivery in dynamic, multi-agent wireless networks. Its practical applications include improving efficiency and user experience in future communication systems, particularly in scenarios requiring reduced traffic loads, lower delivery delays, and efficient spectrum utilization in cognitive radio environments. This could benefit applications like edge AI, smart healthcare, and autonomous driving where timely content access and efficient resource management are critical.

8. Graphs and Figures Analysis

Figure 1: System model

This figure illustrates the system architecture and the multi-agent learning framework. It shows a Primary Base Station (PBS) and multiple Secondary Base Stations (SBSs) acting as edge cache servers. Users access content primarily from SBSs, and secondarily from the PBS if content is not available or fully cached at SBSs. The right side depicts the multi-agent DRL architecture, with K user agents and N SBS agents. Each agent has an Actor network (π) and a Critic network (Q). During training, a centralized critic uses global state (S_All) and global action (A_All) to evaluate actions, while actors generate actions based on their local observations. During execution, agents act decentralized. The red lines indicate centralized training data flow, and green lines indicate decentralized execution data flow.

Table I: Neural Network Configuration of MADDPG

This table details the architecture of the neural networks used for both user and SBS agents in the MADDPG model. For the Actor network, both user and SBS agents have an input layer of size F (number of files), two hidden layers each with 300 neurons and ReLU activation, and an output layer of size F with Sigmoid activation. For the Critic network, both user and SBS agents have an input layer of size 2(N+K)F (global state and action), two hidden layers each with 1024 neurons and ReLU activation, and an output layer of size 1 (policy gradient value scaler).

Figure 2: Locations and rewards of heterogeneous agents

This figure presents the geographical distribution of network entities and the reward convergence for individual agents. Figure 2(a) shows the 2D positions of the PBS, 5 SBSs, and 3 users, illustrating the network topology where SBSs are closer to users than the PBS. Figure 2(b) plots the individual rewards for User 1, User 2, and User 3 against training episodes, showing that all user rewards converge to stable negative values, indicating successful minimization of delivery delay. Figure 2(c) plots the individual rewards for SBS 1 through SBS 5 against training episodes, demonstrating that all SBS rewards converge to stable positive values, signifying optimized cache efficiency and revenue. SBS2 and SBS4 show slightly lower rewards, correlating with their farther locations from users.

Figure 3: Contents popularity vs. cached fraction

This figure analyzes the relationship between content popularity and cached fraction under different conditions. Figure 3(a) illustrates the impact of SBS location on caching strategy, showing the average cached fraction (λ) versus normalized popularity for each SBS. SBSs farther from users (SBS2, SBS4) tend to cache a larger fraction of highly popular content to compensate for worse transmission rates, while closer SBSs exhibit a more balanced caching strategy. Figure 3(b) demonstrates the impact of the replacement cost factor (β) on the caching strategy. As β increases from 0.5 to 1.2, the popularity-fraction curves shift from up-concave to down-convex, indicating that SBSs prioritize caching more popular contents when cache space is limited or replacement costs are higher.

Figure 4: Comparison with homogeneous MADDPG

This figure compares the overall rewards of users and SBSs under different MADDPG configurations over training episodes. Figure 4(a) shows the sum of user rewards, where the proposed "Heter-MADDPG (users and SBSs)" (blue line) achieves the lowest (most negative) reward, indicating superior performance in minimizing user delivery delay compared to homogeneous MADDPG variants. Figure 4(b) shows the sum of SBS rewards, where the "Heter-MADDPG (users and SBSs)" (blue line) achieves the highest positive reward, demonstrating superior performance in maximizing SBS cache efficiency and revenue. The overall conclusion is that the heterogeneous MADDPG significantly outperforms homogeneous approaches in both user delay minimization and SBS cache efficiency maximization.

Figure 5: Performance comparison with other DRL

This figure compares the total global reward (sum of user and SBS rewards) of the proposed heterogeneous MADDPG with other actor-critic based off-policy DRL algorithms: DDPG, PPO, and TD3, over training episodes. The MADDPG (blue line) consistently achieves the highest global reward, converging to a value significantly higher than DDPG, PPO, and TD3. This demonstrates the superior performance of the proposed heterogeneous MADDPG in optimizing the overall system reward by balancing user delivery delay and SBS cache efficiency. The other DRL algorithms show lower and less stable convergence.